

A PROJECT TITLE

ON

Explainable Hybrid Machine Learning Framework for Zero-Day Attack Detection

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

In

A.Y.: 2026-27

By

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CSE



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DEPARTMENT OF CSE

Name of Title: Explainable Hybrid Machine Learning Framework for Zero-Day Attack Detection

1. Abstract:

Zero-day attacks are difficult to detect because their patterns may not be available in the data used for model training. This project proposes an explainable hybrid machine learning framework for detecting known and previously unseen network attacks using the UNSW-NB15 dataset. Random Forest and XGBoost are used to learn known attack patterns, while an autoencoder learns normal network behaviour to identify unusual traffic. The outputs of these models are combined using a hybrid fusion approach to improve detection of unseen attacks. Selected attack categories are excluded from training and used during evaluation to simulate a zero-day detection scenario. The framework is evaluated using precision, recall, F1-score and false-positive rate. SHAP is used to identify important network features contributing to model predictions, improving the interpretability of the detection system. The proposed framework aims to provide a reliable and interpretable approach for detecting both known and previously unseen network attacks.

2. INTRODUCTION:

Network attacks are becoming increasingly difficult to detect, especially when new attack patterns are not present in the training data. Traditional machine learning models mainly detect known attacks, while anomaly detection can identify unusual network behaviour. This project combines both approaches using Random Forest, XGBoost, and an autoencoder to improve zero-day attack detection. The framework also uses SHAP to provide explanations for its predictions, making the detection process more interpretable.

3. SOFTWARE REQUIREMENTS: Python 3.12, Jupyter Notebook, VS Code, Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, SHAP, Matplotlib / Seaborn.

Signature of the Guide

Course Instructor: Nirmalajyothi Nariseti